
Fountain View Hall

Release 0.2.0

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Fountain View Hall's management tool, `fvh-manager` is used to manage reservations at Fountain View Hall.

USER GUIDE

This guide will walk you through `fvh-manager` and covers topics such as how to install and use `fvh-manager`.

1.1 Installation

1.1.1 User Prerequisites

To verify you meet all prerequisites, run:

```
uv --version
```

Expected output:

```
uv 0.x.x
```

Not:

```
command not found: uv
```

Important

Make sure the `uv` command runs before running the program.

If `uv --version` fails, use [this guide](#) to install `uv`.

1.1.2 Minimal Getting Started

Make sure you run these commands from the project directory

Run this once:

```
uv sync
```

Then run this to start on every subsequent run:

```
uv run fvh-manager
```

1.1.3 Activate Completion by Operating System and Shell

If desired, you can activate shell completions by running one of the following commands, depending on your operating system and shell:

Operating system	Shell	Activation command
Linux/macOS	Bash	<code>source completions/fvh-manager.bash</code>
Linux/macOS	Zsh	<code>source completions/_fvh-manager</code>
Linux/macOS	Fish	<code>source completions/fvh-manager.fish</code>
Windows	PowerShell	<code>. .\completions\fvh-manager.ps1</code>

Once shell completions are activated, make sure your virtual environment is active. You can then run the program with:

```
fvh-manager
```

To view available completions, type:

```
fvh-manager <tab>
```

1.2 Developer Installation

Meet all previous *prerequisites* and additional ones as specified bellow.

Install developer dependencies for python:

```
uv sync --all-groups
```

Verify that `pdflatex` is installed:

```
pdflatex -v
```

Verify that `git` is installed:

```
git -v
```

Verify that `lefthook` is installed:

```
lefthook -v
```


REFERENCES

Command-line entry point for the project.

author

Derek R. Neilson

This module defines the project's Typer application and should remain lightweight by design.

Typer is a command-line interface (CLI) framework used throughout this project. It provides decorators such as `typer.command()` for registering CLI commands.

See also

[Typer documentation](#)

`src.main.create_engine(in_memory)`

Create a SQLite database engine.

Creates either an in-memory SQLite database or, by default, a persistent database at `DB_PATH`, which points to `database/data.db`.

Parameters

`in_memory` (bool) – If `True`, create an in-memory database. If `False`, create a persistent database at `DB_PATH`.

Returns

A SQLAlchemy engine representing the configured database connection.

Return type

Engine

The SQLAlchemy database URL has the general form:

```
dialect[+driver]://user:password@host/dbname[?key=value...]
```

The default project database URL has the form:

```
sqlite+pysqlite:///database/data.db
```

If `in_memory` is `True`, the database URL has the form:

```
sqlite+pysqlite:///memory:
```

See also[SQLAlchemy Engine documentation](#) [Typer documentation](#)

```
src.main.start_app(in_memory=False)
```

Start the CLI app.

Parameters

in_memory (bool) – If True, use an in-memory SQLite database.

Return type

None

Sets up logging and metrics for the application.

author

Derek R. Neilson

var logger

Application logger configured for OpenTelemetry.

var meter

OpenTelemetry meter used to create counters and histograms.

This module configures OpenTelemetry logging and metrics, then exposes the `track_command` decorator for tracking CLI command execution.

```
src.observability.configure_logging()
```

Configure and return the application logger.

Return type

Logger

```
src.observability.configure_metrics()
```

Configure and return the application metrics meter.

Return type

Meter

```
src.observability.track_command(command_name)
```

Track command execution with logs and metrics.

The wrapped function records the number of command runs, active operations, errors, and command duration.

Parameters

command_name (str) – Name of the command being tracked. Format not required but should be name-of-command

Return type

Callable[[TypeVar(F, bound= Callable[... Any]), TypeVar(F, bound= Callable[... Any])]

Example

```
from src.observability import logger, track_command

@track_command("some-command")
def some_command():
    logger.info("Command executed")
```

The design process has been documented quite thoroughly, but not all decisions are documented.

3.1 Fountain View Hall Pseudocode and User Journeys

3.1.1 Problem Statement

The project must meet all *requirements*. Besides those requirements, I have set some personal goals as well:

1. Achieve 100% test coverage.

I have done expensive research¹ on this and have concluded that it does not guarantee code standards, but it also does not hurt code quality and is probably to make you think more about your code.

Limitation of Testing

“Program testing can be used to show the presence of bugs, but never to show their absence.”

– Edsger W. Dijkstra [Dijkstra1970]

2. Learn more about Grafana and OpenTelemetry Protocol.

3.2 Overview for Fountain View Hall

3.2.1 Overview

Fountain View Hall wants a program that tracks event attendance and projected revenue for multiple bookings.

3.2.1.1 Program Requirements

- Repeatedly prompt for event name, guest count, and estimated revenue
- Continue until the user enters “done”
- Track total events processed
- Track total guests
- Track total projected revenue
- Calculate the average guests per event
- Determine the largest event

¹ See *Software Testing and Code Coverage* for a list of sources viewed to come to this conclusion.

3.2.1.2 Business Rules

- Pricing Rules
 - Large events receive a discount
 - Operational Rules
 - Every booking must include a guest count
 - Booking total must be calculated before the discount is applied
 - Revenue totals should be tracked
-

3.2.2 Planned Technology Usage

I intended to use various technologies throughout this project; some are very necessary, and others save time. Each technology will be categorized and described below:

3.2.2.1 Editing/Developer Tools

- *Git* and *GitHub* as a VCS (version control system).¹
- *uv* for package, project, and environment management.
- *lefthook* as a Git hook manager.
- *pre-commit* to run pre-commit scripts.²
 - *ruff* for Python linting and formatting running as a pre-commit script.
- *Sphinx* to help build this beautiful documentation.
- *pytest* as a testing suite.
- *Open Telemetry Protocol* for observability and log management.⁴

3.2.2.2 Python Packages

- *typer* for command line interface completions and argument parsing.
 - *SQLite3* as a database³
 - *SQLAlchemy* for object-relational mapping (ORM)
-

3.3 Sources

A curated source list for reStructuredText, OpenTelemetry, SQLAlchemy, and Python tooling.

¹ Go to the [project repo](#) to see commit history.

² All scripts (in the scripts directory) are written for Unix systems and use the bash language.

⁴ You can visit their [web page](#) for more details, but essentially it is a web interface that helps you view logs and trends in data. Probably unnecessary, but I was eager to learn how to use this.

³ See SQLite's [home page](#) for more details.

3.3.1 Quick Reference

3.3.2 reStructuredText and Sphinx

#	Source	Creator	Date	Link
1	reStructuredText Markup Specification	Goodger, David	2026	Open source
2	Restructuredtext Primer - Material for Sphinx	Sheppard, Kevin	8 June 2026	Open source
3	reStructuredText Primer - Sphinx Documentation	Sphinx Documenta- tion	8 June 2026	Open source
4	Welcome to the reStructuredText Primer!	reStructuredText Primer 0.0.1 Docu- mentation	8 June 2026	Open source

3.3.3 OpenTelemetry and Observability

#	Source	Creator	Date	Link
5	How to Set Up a Complete Observability Stack with OpenTelemetry	OneUptime One Complete Observ- ability Platform	6 Feb. 2026	Open source
6	OpenTelemetry	OpenTelemetry	8 June 2026	Open source
7	open-telemetry/opentelemetry- proto	OpenTelemetry - CNCF	22 May 2019	Open source
8	Python OpenTelemetry	OpenTelemetry	8 June 2026	Open source
9	Prometheus Docs	The Linux Founda- tion	9 June 2026	Open source

3.3.4 Python Libraries and Tooling

#	Source	Creator	Date	Link
10	SQLAlchemy	SQLAlchemy	8 June 2026	Open source
11	SQLAlchemy ORM - SQLAlchemy 2.0 Documentation	SQLAlchemy Docu- mentation	8 June 2026	Open source
12	Typer	Typer	8 June 2026	Open source
13	SQLAlchemy Crash Course - Mas- ter Databases in Python	Tim Ruscica	Mar 17, 2025	Open source
14	socket — Low-level networking in- terface	Python Software Foundation	Jun 10, 2026	Open source
15	Python Requests post() Method	W3Schools	2026	Open source

3.3.5 Software Testing and Code Coverage

#	Source	Creator	Date	Link
16	100% Test Coverage is a LIE, Here's Why...	Sam Newman	8 June 2026	Open source
17	Why 100% Test Coverage Isn't the Goal	Daniel Mauno Pettersson	March 4, 2026	Open source
18	Test Coverage	Fowler, Martin	17 Apr. 2012	Open source
19	Coverage Is Not Strongly Correlated with Test Suite Effectiveness	Inozemtseva, Laura; Holmes, Reid	2014	Open source
20	Code Coverage and Postrelease Defects: A Large-Scale Study on Open Source Projects	Kochhar, Pavneet Singh; Lo, David; Lawall, Julia; Nagappan, Nachiappan	2017	Open source
21	A Brief Survey on Oracle-based Test Adequacy Metrics	Hossain, Soneya Binta; Dwyer, Matthew B.	2022	Open source
22	Four Common Types of Code Coverage	web.dev	2024	Open source
23	Getting 100% Code Coverage Doesn't Eliminate Bugs	Kapelonis, Kostis	2017	Open source

3.3.6 Logging

#	Source	Creator	Date	Link
24	Python Logging: Master the Module in Under 10 Minutes	Tim van Cann	Aug 22, 2024	Open source
25	Python JSON Logger	Nicholas Hairs	10 June 2026	Open source
26	Logging in Python	GeeksforGeeks	10 June 2026	Open source

3.3.7 Detailed Source Cards

1. reStructuredText Markup Specification

Type

Specification

Author

Goodger, David

Organization

Docutils

Date

2026

Accessed

7 June 2026

URL
<https://docutils.sourceforge.io/docs/ref/rst/restructuredtext.html>

2. How to Set Up a Complete Observability Stack with OpenTelemetry

Type

Article

Organization

OneUptime | One Complete Observability Platform

Date

6 Feb. 2026

Accessed

7 June 2026

URL<https://oneuptime.com/blog/post/2026-02-06-complete-observability-stack-opentelemetry-open-source/view>

3. Restructuredtext Primer - Material for Sphinx

Type

Documentation

Author

Sheppard, Kevin

Accessed

8 June 2026

URL<https://bashtage.github.io/sphinx-material/basics.html>

4. OpenTelemetry

Type

Website

Organization

OpenTelemetry

Accessed

8 June 2026

URL<https://opentelemetry.io/>

5. open-telemetry/opentelemetry-proto

Type

GitHub Repository

Organization

OpenTelemetry - CNCF

Date

22 May 2019

Accessed

8 June 2026

URL

<https://github.com/open-telemetry/opentelemetry-proto>

Note

Makefile referenced.

6. Python OpenTelemetry

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Organization

OpenTelemetry

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8 June 2026

URL

<https://opentelemetry.io/docs/languages/python/>

7. reStructuredText Primer - Sphinx Documentation

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Accessed

8 June 2026

URL

<https://www.sphinx-doc.org/en/master/usage/restructuredtext/basics.html>

8. SQLAlchemy

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Website

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SQLAlchemy

Accessed

8 June 2026

URL

<https://www.sqlalchemy.org>

9. SQLAlchemy ORM - SQLAlchemy 2.0 Documentation**Type**

Documentation

Organization

SQLAlchemy Documentation

Accessed

8 June 2026

URL<https://docs.sqlalchemy.org/en/20/orm/index.html>**10. Typer****Type**

Documentation

Organization

Typer

Accessed

8 June 2026

URL<https://typer.tiangolo.com/>**11. Welcome to the reStructuredText Primer!****Type**

Documentation

Organization

reStructuredText Primer 0.0.1 Documentation

Accessed

8 June 2026

URL<https://djdeangelis.github.io/reST-primer/index.html>**12. 100% Test Coverage is a LIE, Here's Why...****Type**

Video

Author

Sam Newman

Organization

Modern Software Engineering

Accessed

8 June 2026

URL

<https://www.youtube.com/watch?v=p1xZ-Ni2t8Q>

13. SQLAlchemy Crash Course - Master Databases in Python

Type

Video

Author

Tim Ruscica

Organization

NeuralNine

Date

Mar 17, 2025

Accessed

9 June 2026

URL

<https://www.youtube.com/watch?v=529LYDgRTgQ&t>

14. Prometheus Docs

Type

Documentation

Organization

The Linux Foundation

Accessed

9 June 2026

URL

<https://prometheus.io/docs/introduction/overview/>

15. Why 100% Test Coverage Isn't the Goal

Author

Daniel Mauno Pettersson

Organization

QA.tech

Date

March 4, 2026

Accessed

9 June 2026

URL

<https://qa.tech/blog/why-100-test-coverage-isnt-the-goal>

16. Test Coverage

Type

Article

Author

Fowler, Martin

Organization

Martin Fowler

Date

17 Apr. 2012

Accessed

9 June 2026

URL

<https://martinfowler.com/bliki/TestCoverage.html>

17. Coverage Is Not Strongly Correlated with Test Suite Effectiveness

Type

Conference Paper

Author

Inozemtseva, Laura; Holmes, Reid

Organization

ICSE 2014 / ACM

Date

2014

Accessed

9 June 2026

URL

<https://dl.acm.org/doi/10.1145/2568225.2568271>

18. Code Coverage and Postrelease Defects: A Large-Scale Study on Open Source Projects

Type

Journal Article

Author

Kochhar, Pavneet Singh; Lo, David; Lawall, Julia; Nagappan, Nachiappan

Organization

IEEE Transactions on Reliability

Date

2017

Accessed

9 June 2026

URL

https://ink.library.smu.edu.sg/sis_research/3838/

19. A Brief Survey on Oracle-based Test Adequacy Metrics

Type

Research Survey

Author

Hossain, Soneya Binta; Dwyer, Matthew B.

Organization

arXiv

Date

2022

Accessed

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URL

<https://arxiv.org/abs/2212.06118>

20. Four Common Types of Code Coverage

Type

Technical Article

Organization

web.dev

Date

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URL

<https://web.dev/articles/ta-code-coverage>

21. Getting 100% Code Coverage Doesn't Eliminate Bugs

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Author

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Date

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<https://blog.codepipes.com/testing/code-coverage.html>

22. socket — Low-level networking interface**Type**

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Python Software Foundation

Date

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URL

<https://docs.python.org/3/library/socket.html>

23. Python Requests post() Method**Type**

Website

Organization

W3Schools

Date

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URL

https://www.w3schools.com/PYTHON/ref_requests_post.asp

24. Python Logging: Master the Module in Under 10 Minutes**Type**

Video

Author

Tim van Cann

Organization

Timnology

Date

Aug 22, 2024

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10 June 2026

URL

<https://www.youtube.com/watch?v=A3FkYRN9qog>

Note

Was very helpful.

25. Python JSON Logger

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Author

Nicholas Hairs

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Accessed

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26. Logging in Python

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Accessed

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URL

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26. GeeksforGeeks. *Logging in Python*. Accessed 10 June 2026. <https://www.geeksforgeeks.org/python/logging-in-python/>

3.4 Style Guide

This style guide is for me and serves as a reminder to me.

3.4.1 Scope of Guide

Seeing as I am the sole developer of this project, I don't think it is necessary to define all stylings. That being said, I will define some style choices.

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PYTHON MODULE INDEX

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